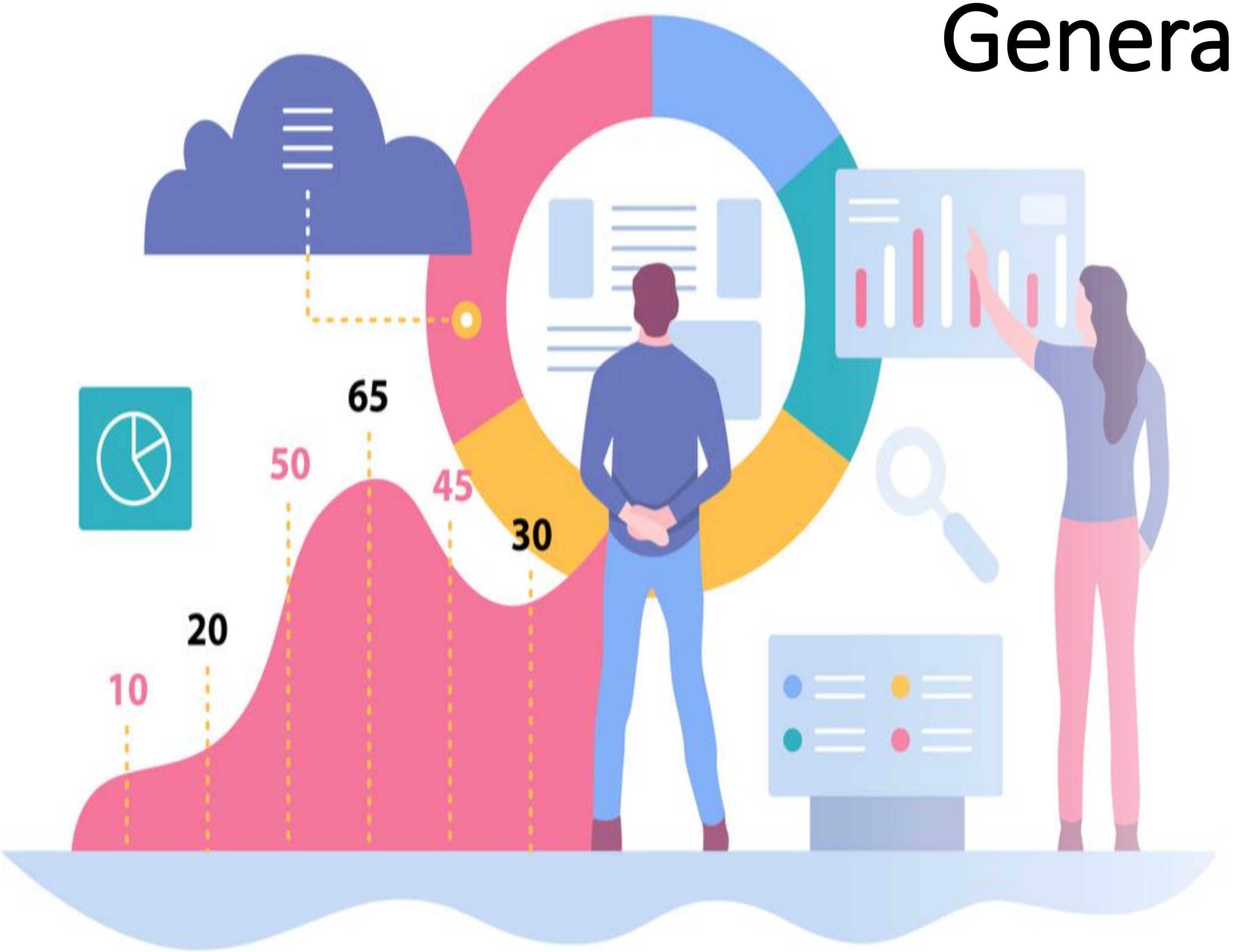
Generalized Linear Models (GLM)



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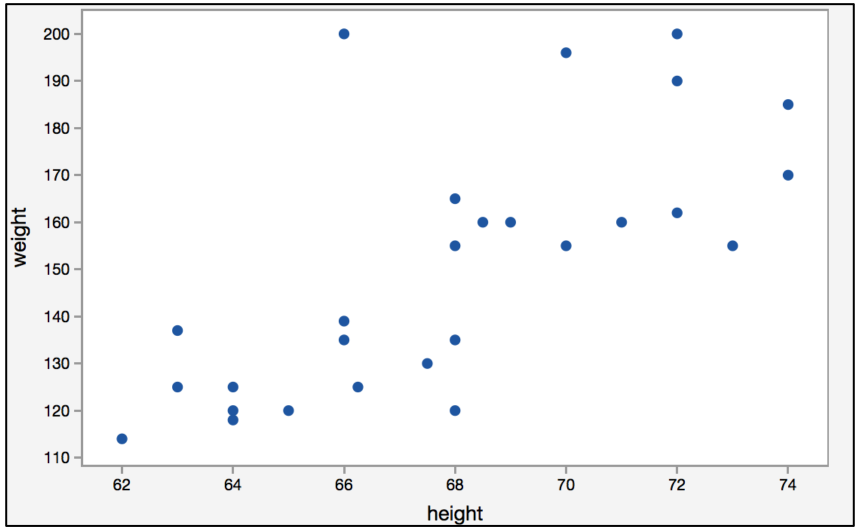
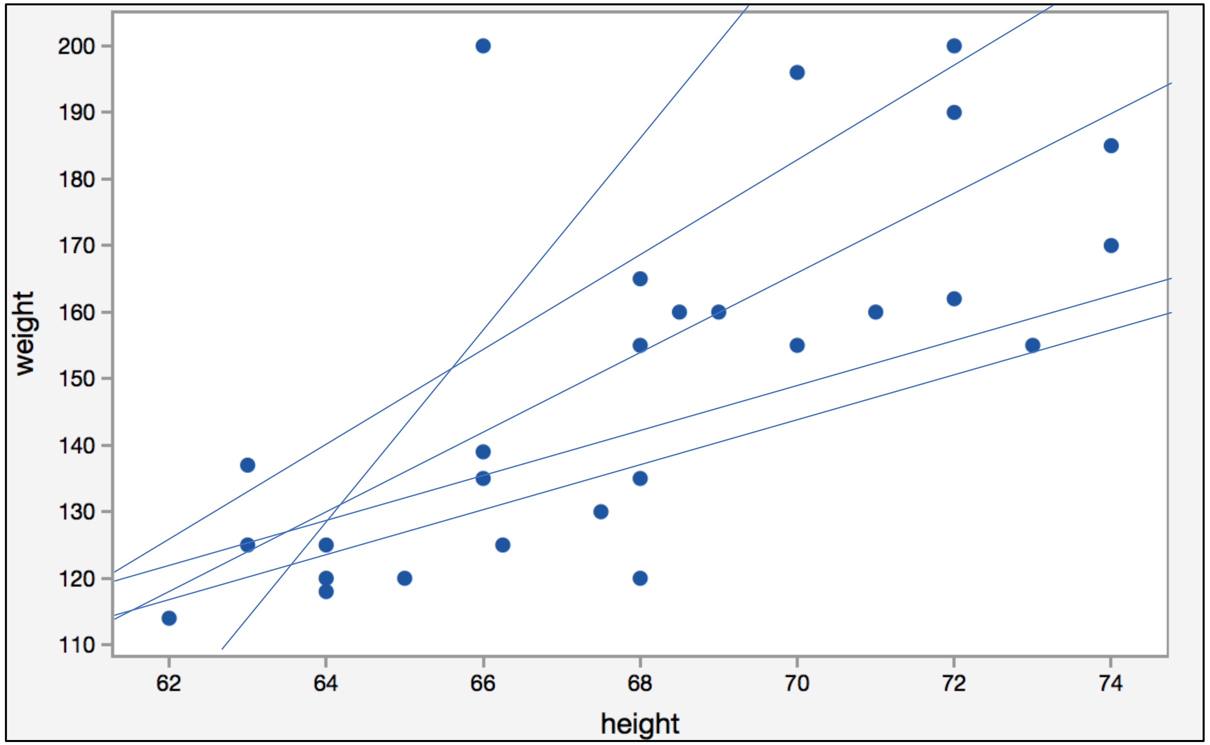
Simple Linear Regression

- Simple linear regression is useful for finding relationship between two variables.
- One is predictor or independent variable and other is response or dependent variable which is a quantitative variable.
- For example, relationship between height and weight.



Simple Linear Regression

- Infinitely many number of plots can be created.



Simple Linear Regression

- The equation for this model for the population is as follows,

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- The values β_0 and β_1 must be chosen so that they minimize the error.
- If sum of squared error is taken as a metric to evaluate the model, then goal to obtain a line that best estimated model which reduces the error.

$$Sum\ of\ Squares\ of\ Error\ (SSE) = \sum_{i=1}^{n} (Actual\ Output - Predicted\ Output)^2$$

$$= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Simple Linear Regression

- This is called as Residual Sum of Squares (RSS) as well. By minimizing this SSE, we can obtain following parameter estimations.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\,(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

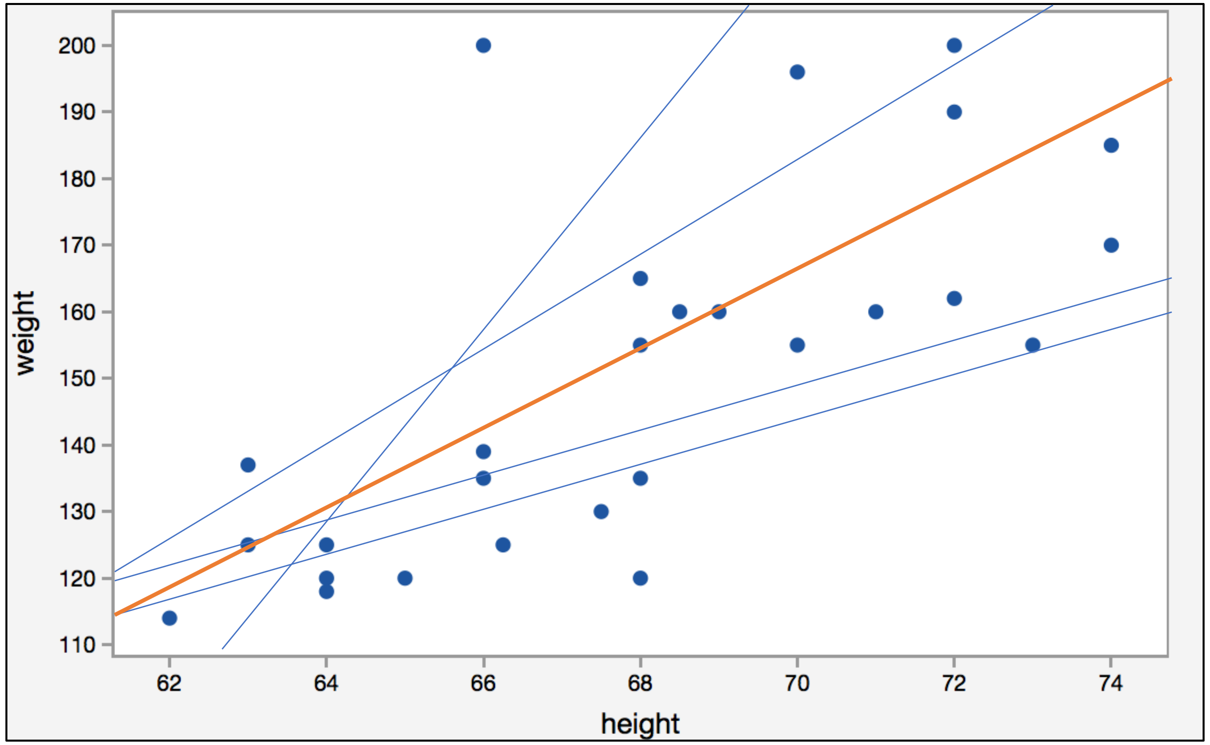
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

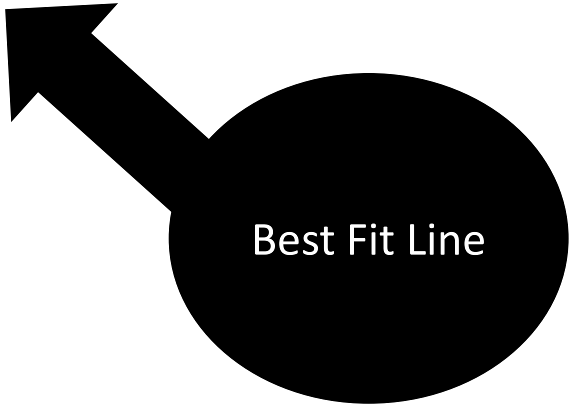
- Then the estimated model is,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Simple Linear Regression

- The core idea is to obtain a line that best fits the data. The best fit line is the one for which total prediction error (all data points) are as small as possible.





Multiple Linear Regression

- The equation for this model is as follows,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_k x_k + \varepsilon$$

- Consider here we have k variables.
- By minimizing this SSE, we can obtain the parameter estimations in here as well.

Linear Regression Assumptions

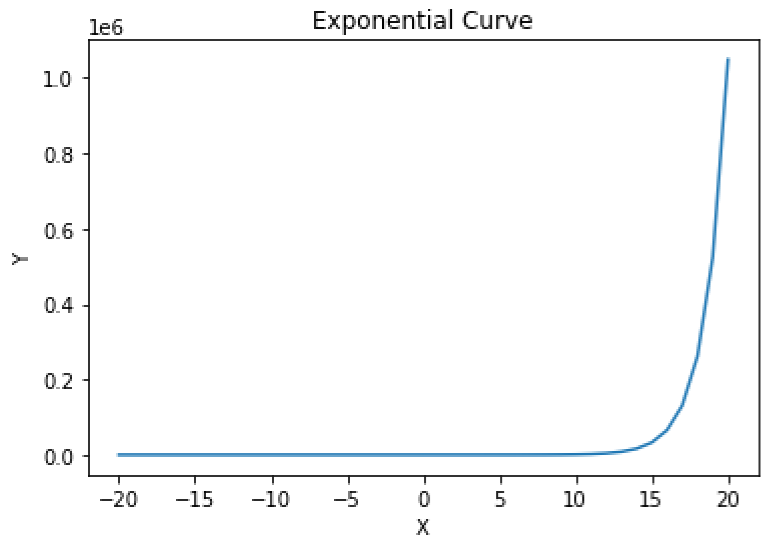
There are mainly five assumptions associated with a linear regression model:

- **Linearity**: The relationship between X and the mean of Y is linear.
- **Multicollinearity**: Correlation among independent variables.
- **Homoscedasticity**: The variance of residual is the same for any value of X.
- **Independence**: Observations are independent of each other.
- **Normality**: For any fixed value of X, Y is normally distributed.

Linear Regression Issues

Linear Regression model is not suitable if,

- The relationship between X and y is not linear. There exists some non-linear relationship between them. For example, y increases exponentially as X increases.



Linear Regression Issues

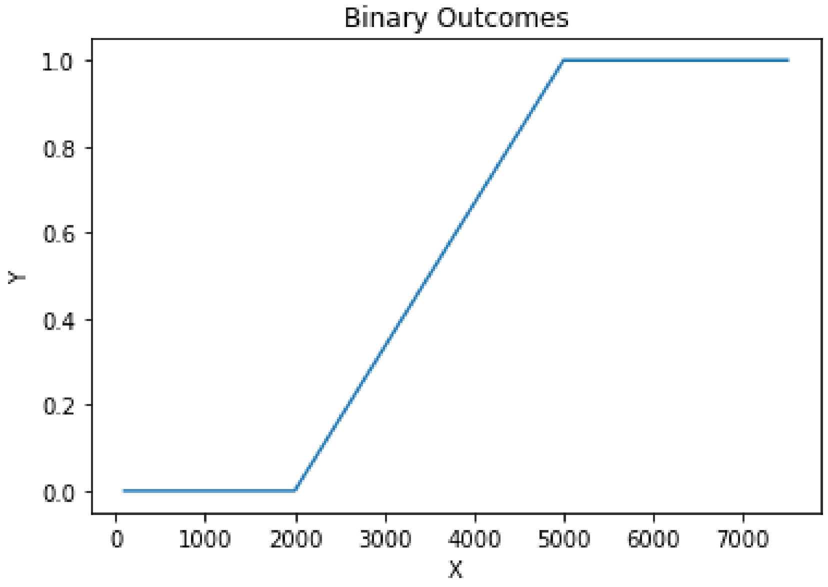
Linear Regression model is not suitable if,

- Variance of errors in y (commonly called as Homoscedasticity in Linear Regression), is not constant, and varies with X.
- Response variable is not continuous, but discrete/categorical. Linear Regression assumes normal distribution of the response variable, which can only be applied on a continuous data.
- If we try to build a linear regression model on a discrete/binary y variable, then the linear regression model predicts negative values for the corresponding response variable, which is inappropriate.

Linear Regression Issues

Consider the following example.

- In the below graph, we can see the response is either 0 or 1.
- When X <5000, y is 0, and when X≥5000, y is 1.



Linear Regression Issues

For Example, Consider a linear model as follows:

A simple example of a mobile price in an e-commerce platform:

$$\text{Price} = 12500 + 1.5 \times \text{Screen size} - 3 \times \text{Battery Backup (less than 4hrs)}$$

Data available for,

- Price of the mobile
- Screen size (in inches)
- Is battery backup less than 4hrs – with values either as ‘yes’, or ‘no’.

Linear Regression Issues

- In this example, if the screen size increases by 1 unit, then the price of the mobile increases by 1.5 times the default price, keeping the intercept (12500) and Battery Backup values constant.
- Likewise, if the Battery Backup of less than 4hrs is ‘yes, then the mobile price reduces by three times the default price.
- If the Battery Backup of less than 4hrs is ‘no’, then the mobile price is unaffected, as the term ($3 \times$ Battery Backup) becomes 0 in the linear model.
- The intercept 12500 indicates the default price for a standard value of screen size.
- This is a valid model.

Linear Regression Issues

- However, if we get a model as below:

$$\text{Price} = 12500 + 1.5 \times \text{Screen size} + 3 \times \text{Battery Backup(less than 4hrs)}$$

- Here, if the battery backup less than 4 hrs is ‘yes, then the model is saying the price of the phone increases by three times.
- Clearly, from practical knowledge, we know this is incorrect.
- There will be less demand for such mobiles. These are going to be very old mobiles, which when compared to the current range of mobiles with the latest features, is going to be very less in price.
- This is because the relationship between the two variables is not linear, but we are trying to express it as a linear relationship. Hence, an invalid model is built.

Linear Regression Issues

- Similarly, if we are trying to predict if a particular phone will be sold or not, using the same independent variables, but the target is we are trying to predict if the phone will sell or not, so it has only binary outcomes.
- Using Linear Regression, we get a model like,

$$\text{Sales} = 12500 + 1.5 \times \text{Screen size} - 3 \times \text{Battery Backup(less than 4hrs)}$$

Linear Regression Issues

- This model doesn't tell us if the mobile will be sold or not, because the output of a linear regression model is continuous value. It is possible to get negative values as well as the output.
- It does not translate to our actual objective of whether phones having some specifications based on the predictors, will sell or not (binary outcome).
- Similarly if we are also trying to see what is the number of sales of this mobile that will happen in the next month, a negative value means nothing.
- Here, the minimum value is 0 (no sale happened), or a positive value corresponding to the count of the sales.
- Having the count as a negative value is not meaningful to us.

What is GLM

- Generalized Linear Model (GLM) is an advanced statistical modelling technique formulated by John Nelder and Robert Wedderburn in 1972.
- It is an umbrella term that encompasses many other models, which allows the response variable y to have an error distribution other than a normal distribution.
- The models include Linear Regression, Logistic Regression, and Poisson Regression.

Assumptions of GLM

- Similar to Linear Regression Model, there are some basic assumptions for Generalized Linear Models as well.
- Most of the assumptions are similar to Linear Regression models, while some of the assumptions of Linear Regression are modified.

➢ Data should be independent and random (Each Random variable has the same probability distribution).

➢ The response variable y does not need to be normally distributed, but the distribution is from an **exponential family** (e.g. binomial, Poisson, multinomial, normal)

➢ The original response variable need not have a linear relationship with the independent variables, but the transformed response variable (through the link function) is linearly dependent on the independent variables

Assumptions of GLM

- Feature engineering on the Independent variable can be applied i.e instead of taking the original raw independent variables, variable transformation can be done, and the transformed independent variables, such as taking a log transformation, squaring the variables, reciprocal of the variables, can also be used to build the GLM model.
- Homoscedasticity (i.e constant variance) need not be satisfied.
  - Response variable Error variance can increase, or decrease with the independent variables.
- Errors are independent but need not be normally distributed

Exponential Family of Distributions

- The exponential family of distribution is the set of distributions parametrized by $\theta \in \mathbf{R}^D$
- That can be described in the form:

$$p(x \mid \theta) = h(x) \exp\left(\eta(\theta)^T T(x) - A(\theta)\right)$$

Example: Poison Distribution

$X \sim Poisson(\theta)$, where $E[X] = \theta$.

$$
\begin{aligned}
p(x \mid \theta) &= \frac{\theta^x}{x!} e^{-\theta}, \quad x = 0, 1, \ldots \\
&= \frac{1}{x!} exp\{(log(\theta)x - \theta\} \\
&= h(x) exp\{\eta(\theta) T(x) - B(\theta)\}
\end{aligned}
$$

where:

- $h(x) = \frac{1}{x!}$
- $\eta(\theta) = log(\theta)$
- $T(x) = x$

Example: Binomial Distribution

$$X \sim Binomial(\theta, n)$$

$$\begin{aligned} p(x \mid \theta) &= \binom{n}{x} \theta^x (1-\theta)^{n-x} \quad x = 0, 1, \ldots, n \\ &= \binom{n}{x} exp\{log(\tfrac{\theta}{1-\theta})x + nlog(1-\theta)\} \\ &= h(x) exp\{\eta(\theta) T(x) - B(\theta)\} \end{aligned}$$

where:

- $h(x) = \binom{n}{x}$
- $\eta(\theta) = log(\frac{\theta}{1-\theta})$
- $T(x) = x$
- $B(\theta) = -nlog(1-\theta)$

Example: Normal Distribution

$X \sim N(\mu, \sigma_0^2)$. (Known variance)

$$\begin{aligned}
p(x \mid \theta) &= \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x-\mu)^2} \\
&= [\frac{1}{\sqrt{2\pi\sigma_0^2}} exp\{-\frac{x^2}{2\sigma_0^2}\}] \times exp\{\frac{\mu}{\sigma_0^2}x - \frac{\mu^2}{2\sigma_0^2}\} \\
&= h(x) exp\{\eta(\theta)T(x) - B(\theta)
\end{aligned}$$

where:

- $h(x) = [-\frac{1}{\sqrt{2\pi\sigma_0^2}} exp\{-\frac{x^2}{2\sigma_0^2}\}]$
- $\eta(\theta) = \frac{\mu}{\sigma_0^2}$
- $T(x) = x$
- $B(\theta) = \frac{\mu^2}{2\sigma_0^2}$

Example: Normal Distribution

$X \sim N(\mu_0, \sigma^2)$. (Known mean)

$$\begin{aligned}
p(x \mid \theta) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu_0)^2} \\
&= [\frac{1}{\sqrt{2\pi\sigma^2}}] \times exp\{-\frac{1}{2\sigma^2}(x-\mu_0)^2) - \frac{1}{2} log(\sigma^2)\} \\
&= h(x) exp\{\eta(\theta) T(x) - B(\theta)
\end{aligned}$$

where:

- $h(x) = [\frac{1}{\sqrt{2\pi}}]$
- $\eta(\theta) = -\frac{1}{2\sigma^2}$
- $T(x) = (x - \mu_0)^2$
- $B(\theta) = \frac{1}{2} log(\sigma^2)$

Back to GLM

- There are 3 components in GLM.

1. Systematic Component/Linear Predictor
2. Link Function
3. Random Component/Probability Distribution

Systematic Component/Linear Predictor

- It is just the linear combination of the Predictors and the regression coefficients.

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_k x_k$$

Link Function

- Represented as η or g(μ), it specifies the link between a random and systematic components.
- It indicates how the expected/predicted value of the response relates to the linear combination of predictor variables.

Random Component/Probability Distribution

- It refers to the probability distribution, from the family of distributions, of the response variable.
- The family of distributions, called an exponential family, includes normal distribution, binomial distribution, or poisson distribution.

Examples GLM – Linear Regression

- Linear Regression, for continuous outcomes with normal distribution.
- Here we model the mean expected value of a continuous response variable as a function of the explanatory variables.
- Identity link function is used, which is the simplest link function.
- If there is only 1 predictor, then the model is called Simple Linear Regression.
- If there are 2 or more explanatory variables, then the model is called Multiple Linear Regression.

$$y = \beta_0 + \beta_1 x_1 \qquad\qquad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_k x_k$$

- Response is continuous
- Predictors can be continuous or categorical, and can also be transformed.
- Errors are distributed normally and variance is constant.

Examples GLM – Binary Logistic Regression

- For dichotomous or binary outcomes with binomial distribution.
- Here Log odds is expressed as a linear combination of the explanatory variables.
- Logit is the link function.
- The Logistic or Sigmoid function, returns probability as the output, which varies between 0 and 1

$$log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_p x_p$$

- Response variable has only 2 outcomes
- Predictors can be continuous or categorical, and can also be transformed.

Examples GLM – Poisson Regression

- Here count values are expressed as a linear combination of the explanatory variables.
- Log link is the link function.

$$log(\lambda) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_p x_p$$

- where λ is the average value of the count variable
- Response variable is a count value per unit of time and space
- Predictors can be continuous or categorical, and can also be transformed.